

Kumudini Gandesiri

kumudinigandesiri@gmail.com | +1 (925) 315-1110

Engineering Technician | Cupertino, CA

RF, Engineering, Electromechanical & PCB Systems Laboratory Expert | Industrial & Electronics Lab Expertise

PROFESSIONAL EXPERIENCE

AMD (Advanced Micro Devices)

Oct 2025 – Present

Hardware Lab Technician – x86 & SoC Validation

San Jose, CA

- Operate and maintain engineering validation labs supporting x86 and SoC platforms, ensuring continuous system uptime, connectivity, and lab infrastructure readiness in a high-throughput hardware development environment.
- Perform power testing and electrical characterization using oscilloscopes, DMMs, DAQ systems, and power analyzers, measuring and validating device performance across a range of operating conditions consistent with CMOS and DRAM characterization workflows.
- Execute BIOS configuration and low-level platform bring-up, conducting systematic root cause investigation and debug of hardware failures on SoC validation platforms using FMEA and 8D problem-solving frameworks.
- Develop and maintain Python and shell scripts to automate data collection, measurement workflows, and test reporting, enabling structured trend monitoring aligned with SPC and FDC principles.
- Coordinate process, equipment, and material evaluations during new platform introduction, working with cross-functional teams to validate and qualify hardware configurations and measurement setups.
- Monitor system health metrics and performance trends, identify deviations from baseline, and escalate anomalies to engineering teams to support continuous process improvement and reliability objectives.
- Enforce ESD, lab safety, and cleanroom-equivalent handling protocols for sensitive semiconductor hardware, maintaining compliance with quality and reliability standards.
- Produce structured test reports documenting procedures, measurement results, and debug findings, maintaining full hardware configuration traceability to support engineering review and audit requirements.

Apple Inc

June 2024 – Oct 2025

Test Engineering Technician – Embedded Systems & Hardware Support

Santa Clara, CA

- Collaborated with global teams on silicon validation and reliability analysis from NPI to post-ramp stages.
- Monitored probed materials and yield accounting to optimize process yield and reduce RMAs.
- Hands-on experience with oscilloscopes, DMMs, signal generators, power supplies, network analyzers, spectrum analyzers, and power meters for electrical and RF testing.
- Performed RF testing and signal analysis across various test setups and technologies.
- Skilled in SMD (SMT) and Through-Hole Technology (THT) soldering, including board-level rework, component replacement, and repair.
- Identified signal and system failures through hands-on hardware and system-level debugging using oscilloscopes, differential probes active & passive probes and related test equipment.
- Executed electrical and physical failure analysis using SEM, optical microscope, curve tracer, and multimeter.
- Performed detailed failure analysis, capturing measurements, waveforms, and observations, and documented test results beyond basic pass/fail reporting.
- Conducted first-level triaging, logging results and working closely with part owners and design engineers.
- Automated data parsing with Python and Excel to identify root cause patterns.
- Authored technical investigation reports on failure modes and corrective actions.
- Performed equipment troubleshooting, calibration, and maintenance per lab safety and ESD protocols.
- Contributed to lab efficiency improvements, SOP updates, and new tool evaluations.
- Maintained certifications and knowledge of Apple SoC and device architecture.
- Provided 24/7 support partnering with cross-regional teams during validation cycles.

Team Engineers Pvt Ltd

July 2022 – Jan 2023

Engineering Technician – Embedded Systems & Hardware Support

Hyderabad, India

- Monitored probed material and performed manual yield accounting using defect references.
- Performed electrical and physical failure analysis to identify device and board-level issues.

- Executed basic setup, operation, and troubleshooting of lab equipment, including SEM, curve tracers, and microscopes.
- Hands-on experience with oscilloscopes, DMMs, signal generators, power supplies, network analyzers, spectrum analyzers, and power meters for electrical and RF testing.
- Experienced in precision soldering of ultra-small SMD components (0201 package) using microscope-assisted techniques.
- Skilled in SMD (SMT) and Through-Hole Technology (THT) soldering, including board-level rework, component replacement, and repair.
- Conducted initial data analysis and documented findings to assist part owners and design engineers.
- Generated consolidated investigation reports outlining failure modes and potential root causes.
- Collaborated with engineering teams to support product and process development.
- Maintained compliance with ESD and lab safety protocols; ensured equipment certifications were up to date.
- Contributed to continuous improvement efforts by suggesting workflow and equipment process enhancement.
- Completed and maintained technical training certifications relevant to lab operations.
- Stayed updated on device structures and participated in yield analysis reviews to support ongoing production.

TECHNICAL SKILLS

- **Failure Analysis Tools:** SEM, Curve Tracer, Optical Microscope, DMM, Power Supplies, Signal Generator, Power/Network/Spectrum Analyzer, Power Meter, RF Testing
- **Test & Automation:** Python, PyVisa, Serial, Shell Scripting, Oscilloscope, Logic Analyzers, DAQ Systems
- **Platforms:** Linux, Ubuntu, RHEL, macOS, Windows
- **Lab Tools:** Excel, JMP, Git, JIRA, Confluence
- **Hardware:** Microcontrollers, SoCs, PCBs, Embedded Systems, x86 Platforms, BIOS Configuration
- **Methodologies:** DOE, SPC, FDC, FMEA, 8D Root Cause Analysis
- **Certifications:** Apple Lab Safety, ESD Compliance, Python for Data Analysis

EDUCATION

Master of Science – Management and Information Systems

Aug 2024

Northwest Missouri State University

Maryville, Missouri, USA

- Coursework in Data Analytics, IT Management, and System Integration.

Bachelor of Technology – Electronics and Communication Engineering

June 2022

Jawaharlal Nehru Technological University

Hyderabad, India